Diamond DA42 NG Twin Star (DA42)

2× Austro Engine AE300 turbo-diesel, 168 hp each (FADEC / ECU, single power lever per engine) · 4 seats · Typical DA42 NG (AE300, G1000) AFM procedures; speeds are NG figures and differ from the DA42 TDI (Thielert) – switch labels (ECU A/B voter, ECU TEST) vary slightly by serial.

V-speeds & limits

Vr	**76 KIAS (flaps UP) / 71 KIAS (flaps APP)**
Vx	**84 KIAS**
Vy	**90 KIAS**
Vmca	**68 KIAS**
Vyse	**85 KIAS**
Vxse	**82 KIAS**
Best glide	**88 KIAS (both props feathered, flaps UP)**
Vfe	**137 KIAS (APP) / 113 KIAS (LDG)**
Vle / Vlo	**194 KIAS extended / 156 KIAS retraction**
Va	**125 KIAS**
Vno	**155 KIAS**
Vne	**194 KIAS**
Vs / Vso	**68 / 62 KIAS**
Vref	**86 KIAS (flaps LDG)**

Normal procedures

Before Start

- Preflight inspection — **COMPLETE (fuel drains, oil & coolant levels, gearbox oil, tow bar removed)**
- Passenger briefing — **COMPLETE (belts, canopy / rear door, exits, no smoking)**
- Seats, belts, harnesses — **ADJUST & LOCK**
- Front canopy — **CLOSED & LOCKED (or cooling gap for taxi)**
- Rear door — **CLOSED & LOCKED**
- Parking brake — **SET**
- Gear selector — **DOWN**
- Power levers — **IDLE**
- Fuel selectors — **NORMAL (both)**
- Fuel transfer pumps — **OFF**
- ECU A/B (voter) switches — **AUTO (both)**
- Alternators L / R — **OFF**
- Avionic master — **OFF**
- Electric master — **ON**
- Annunciators / G1000 — **CHECK (GEAR three green, DOOR message off)**
- Fuel quantity — **CHECK mains and aux vs visual check**
- Flaps — **CHECK operation, set UP**
- Strobes / position lights — **ON**

Engine Start

- Propeller area — **CLEAR (start LEFT engine first, then RIGHT)**
- Power lever — **IDLE**
- ENGINE MASTER — **ON (ECU self-test, glow plugs preheat)**
- L / R GLOW annunciation — **WAIT until extinguished**
- Starter — **ENGAGE (max 10 s, then 2 min cooling)**
- Oil pressure — **CHECK rising within 3 s, green within 30 s**
- RPM — **CHECK idle (approx 890 RPM) / engine stable**
- ALTERNATOR — **ON – check charging, no L/R ALTN FAIL**
- Repeat for other engine — **COMPLETE**
- Avionic master — **ON**
- Engine instruments — **CHECK (oil / coolant / gearbox temps rising)**
- Pitot heat — **TEST then OFF**

Before Taxi

Flight instruments / G1000	**CHECK & SET (altimeter, HSI, CDI)**
Radios / transponder	**SET**
Trims (elevator / rudder)	**SET for takeoff**
Flaps	**UP (or APP for short field)**
Parking brake	**RELEASE**
Brakes	**CHECK immediately after moving**

Nosewheel is free-castoring: steer with differential brakes and asymmetric power. Avoid riding the brakes.

Turn coordinator / HSI	**CHECK during turns**

Before Takeoff

Parking brake	**SET**
Flight controls	**FREE & CORRECT**
Trims	**SET takeoff**
Fuel selectors	**NORMAL (both)**
Fuel transfer pumps	**OFF**
Engine temperatures	**OIL min 50°C, COOLANT min 60°C for takeoff**
Ground check – LEFT engine:	**POWER LEVER IDLE**
ECU TEST L button	**PRESS & HOLD – RPM rises, propeller cycles to feather and back, ECU A FAIL / ECU B FAIL appear then clear**
ECU TEST L button	**RELEASE after test complete (no ECU caution remains)**
ECU A/B switch L	**B – engine runs normally; A – engine runs normally; back to AUTO**
Ground check – RIGHT engine:	**REPEAT ECU TEST and ECU A/B check**
Power levers	**IDLE (check RPM stable, no asymmetric indication)**
Engine instruments	**CHECK green, no caution messages**
Electrical	**CHECK voltage / both alternators charging**
Flaps	**UP or APP as planned**
Front canopy	**CLOSED & LOCKED (no gap)**
Rear door	**LOCKED (DOOR annunciation off)**
Pitot heat / lights	**AS REQUIRED**
Transponder	**ALT**
Takeoff briefing	**COMPLETE (Vr 76, Vyse 85, Vmca 68, engine failure plan, gear up at positive rate)**

After Takeoff / Climb

Takeoff power	**CHECK 100% load / 2,300 RPM both engines**
Airspeed	**Vy 90 KIAS (obstacles: Vx 84 KIAS), never below Vyse 85 once gear is up**
Landing gear	**UP (positive rate, below 156 KIAS) – lights out**
Flaps	**UP (if APP used, above 85 KIAS)**
Climb power	**SET (92% load or as required; FADEC manages mixture / prop)**
Engine instruments	**CHECK (oil / coolant / gearbox temps green)**

Cruise

Power levers	**SET (typ. 60–75% load, cruise speed 150–160 KTAS)**
Fuel quantity	**CHECK mains & aux, balance within limits**
Fuel transfer pumps	**ON when main tank below approx 2/3 – transfers aux to main, OFF when complete (L/R FUEL TRANSFER message)**

Do not transfer with a nearly full main tank – surplus fuel vents overboard. Transfer stops when the aux tank is empty.

Engine instruments / ECU	**CHECK (no ECU A / B caution)**
Electrical	**CHECK voltage / load**

Descent

Altimeter	**SET**
Power levers	**REDUCE as required (no shock-cooling limits, FADEC controlled)**
Fuel selectors	**NORMAL**
Fuel transfer pumps	**OFF**
Approach briefing	**COMPLETE (ATIS, approach, speeds, single-engine plan)**

Before Landing

Seats / belts	**SECURE**
Fuel selectors	**NORMAL**
Airspeed	**BELOW 194 KIAS (Vlo extension)**
Landing gear	**DOWN – three green, hydraulic pump off**
Flaps	**APP below 137 KIAS, LDG below 113 KIAS (when landing assured)**
Airspeed	**FINAL 86 KIAS (flaps LDG), short field 82 KIAS**
Landing light	**ON**
Gear	**RE-CHECK three green on final**

After Landing

Flaps	**UP**
Pitot heat	**OFF**
Transponder	**STBY / as required**
Lights	**AS REQUIRED**
Front canopy	**COOLING GAP position permitted for taxi**

Shutdown

Parking brake	**SET**
Power levers	**IDLE, 2 min cool-down (turbo)**
Avionic master	**OFF**
Alternators L / R	**OFF**
ENGINE MASTER L	**OFF**
ENGINE MASTER R	**OFF**
Electric master	**OFF (after propellers stop)**
Fuel selectors	**NORMAL (leave on)**
Canopy	**CLOSED, sun cover installed**
Pitot covers / chocks / tie-downs	**INSTALL**

Abnormal procedures

ECU A or ECU B Failure

L/R ECU A FAIL or L/R ECU B FAIL caution

ECU A/B switch (affected engine)	**CHECK AUTO – the ECU voter has switched to the serviceable unit**
Engine	**CHECK running normally (RPM, load)**
Power lever	**AVOID rapid movements**

▶ Both ECU A and ECU B caution on same engine?

Yes:

ECU A/B switch	**TRY A, then B manually**
Engine failure	**EXPECT – prepare one-engine-inoperative procedure**
Land	**AS SOON AS POSSIBLE**

No – single ECU:

Deferred: One ECU failed – no redundancy on that engine, land as soon as practical. — **NOTED**

Alternator Failure / Low Voltage

L/R ALTN FAIL caution or LOW VOLTS warning

ALTERNATOR (affected side)	**OFF, then ON (reset once)**

▶ Alternator restored?

Yes:

Electrical load	**MONITOR**

No – one alternator:

ALTERNATOR (affected side)	**OFF**
Electrical load	**REDUCE to within single alternator capacity**

Deferred: One alternator failed – single-source electrics, reduced load, land as soon as practical. — **NOTED**

No – both failed / LOW VOLTS:

Both ALTERNATOR switches	**OFF**
Non-essential loads	**OFF (lights, pitot heat if not required, air conditioning)**
Battery	**MONITOR (approx 30 min endurance at reduced load – verify per AFM)**

Engines keep running on ECU backup batteries. Extend gear early while battery power remains, or use emergency extension.

Deferred: On battery only – extend gear early (or free-fall), flaps once on final, land as soon as possible. — **NOTED**

Low Oil Pressure / High Oil Temperature

L/R OIL PRES or L/R OIL TEMP caution

Oil pressure / temperature (affected engine) ..... **CHECK and cross-check**

Power lever (affected engine) ..... **REDUCE**

▶ Pressure below red line or temperature rising above limit?

Yes:

Engine ..... **SHUT DOWN & FEATHER (ENGINE MASTER OFF) before seizure**

Continue per One Engine Inoperative.

Deferred: One engine shut down – single-engine approach, Vyse 85 KIAS. ..... **NOTED**

No – borderline:

Power ..... **MINIMUM practical, monitor**

Deferred: Oil pressure / temperature abnormal – land as soon as practical, be ready to shut down the engine. ..... **NOTED**

High Coolant / Gearbox Temperature

L/R COOLANT TEMP or L/R GBX TEMP caution

Power lever (affected engine) ..... **REDUCE**

Airspeed ..... **INCREASE (more cooling airflow)**

Coolant / gearbox temperature ..... **MONITOR**

▶ Temperature returning to green?

Yes:

Power ..... **KEEP REDUCED, monitor**

Deferred: Engine overheated – reduced power on one engine, maintenance inspection. ..... **NOTED**

No:

Engine ..... **SHUT DOWN & FEATHER (ENGINE MASTER OFF)**

Deferred: One engine shut down for overheat – single-engine approach, Vyse 85 KIAS. ..... **NOTED**

Low Fuel / Fuel Pressure

L/R FUEL LOW or L/R FUEL PRESS caution

Fuel quantity ..... **CHECK main and aux (affected side)**

Fuel transfer pump (affected side) ..... **ON if aux fuel remains**

Fuel selector (affected engine) ..... **X-FEED from other main tank if imbalance / starvation**

▶ Caution extinguished?

Yes:

Fuel ..... **MONITOR, keep crossfeed only as needed**

Deferred: Fuel system abnormal – land as soon as practical. ..... **NOTED**

No:

Engine failure ..... **EXPECT – prepare one-engine-inoperative procedure**

Land ..... **AS SOON AS POSSIBLE**

Gear Fails to Retract / Gear Unsafe Light

Airspeed ..... **BELOW 156 KIAS**

Gear selector ..... **CHECK UP, recycle DOWN then UP once**

GEAR circuit breaker ..... **CHECK IN**

▶ Gear retracted and lights out?

Yes:

Flight ..... **CONTINUE, monitor hydraulic pump cycling**

No:

Gear selector ..... **DOWN – three green**

Airspeed ..... **LIMIT to 194 KIAS, plan fuel for gear-down drag**

Deferred: Gear down and locked – return for landing, do not retract, increased fuel burn. ..... **NOTED**

Canopy or Rear Door Open in Flight

DOOR caution or canopy / door unlatched

Airspeed ..... **REDUCE below 137 KIAS (Vfe APP)**

Canopy / door ..... **DO NOT attempt to close in flight**

The canopy stays in the cooling gap position; the rear door opens a few inches. Expect noise and buffet; the aircraft remains controllable.

Loose objects / belts ..... **SECURE**

Deferred: Canopy / door open – land as soon as practical, keep speed below 137 KIAS. ..... **NOTED**

PFD / AHRS / ADC Failure

Display backup (red button) ..... **PRESS – reversionary mode on MFD**

Standby instruments ..... **USE (airspeed, attitude, altimeter, compass)**

Autopilot ..... **OFF if AHRS failed**

PFD / AHRS / ADC circuit breakers ..... **CHECK**

▶ Flight in IMC?

Yes:

ATC ..... **ADVISE, request radar assistance / nearest VMC**

Deferred: Primary flight display failed – partial panel, request vectors, avoid IMC. ..... **NOTED**

No:

Deferred: PFD / AHRS failed – standby instruments only. ..... **NOTED**

Inadvertent Icing Encounter

LAND AS SOON AS PRACTICAL

Pitot heat ..... **ON**

De-ice system (if fitted) ..... **ON (HIGH / MAX), check fluid**

Exit icing conditions ..... **TURN BACK / CHANGE ALTITUDE**

Power ..... **INCREASE to keep propellers clear**

Autopilot ..... **OFF, monitor control feel**

Landing ..... **FLAPS UP or APP, approach 95–100 KIAS, gear down late**

Deferred: Residual airframe ice – flaps-up / APP landing, approach 95–100 KIAS, longer runway. ..... **NOTED**

Emergency procedures

Engine Failure During Takeoff Roll

Power levers (memory) ..... **BOTH IDLE**

Brakes (memory) ..... **APPLY, maintain direction with rudder / differential brake**

ENGINE MASTERS ..... **OFF (both) when stopped**

Fuel selectors ..... **OFF**

Electric master ..... **OFF**

Engine Failure after Takeoff (gear down / low altitude)

Airborne, insufficient runway remaining

LAND IMMEDIATELY

Power levers (memory) ..... **BOTH MAX**

Airspeed (memory) ..... **85 KIAS (Vyse) – never below 68 KIAS (Vmca)**

Landing gear (memory) ..... **UP (if climb required)**

Flaps (memory) ..... **UP**

Failed engine (memory) ..... **IDENTIFY (dead foot – dead engine), VERIFY with power lever**

ENGINE MASTER (failed engine) (memory) ..... **OFF – propeller feathers automatically**

Bank approx 5° into the operating engine, ball half out. If a climb at Vyse cannot be achieved or runway remains, land straight ahead: gear DOWN, flaps as required, both ENGINE MASTERs OFF before touchdown.

Feathering ..... **CONFIRM (propeller stopped / low RPM); if not feathered expect reduced climb**

ALTERNATOR (failed engine) ..... **OFF**

Fuel selector (failed engine) ..... **OFF**

Return ..... **ONE-ENGINE-INOPERATIVE APPROACH**

Engine Failure in Flight / One Engine Inoperative

Loss of power on one engine with altitude available

LAND AS SOON AS POSSIBLE

- Power levers (memory) — **BOTH MAX (as required)**
- Airspeed (memory) — **85 KIAS (Vyse) or higher**
- Landing gear / flaps (memory) — **UP**
- Failed engine (memory) — **IDENTIFY (dead foot – dead engine), VERIFY with power lever**

▶ Time and altitude for a restart attempt?

Yes – restart:

- ECU A/B switch (failed engine) — **CHECK AUTO, try B then A**
- Fuel selector (failed engine) — **X-FEED (if fuel starvation suspected), fuel quantity check**
- Power lever (failed engine) — **IDLE**
- ENGINE MASTER (failed engine) — **OFF, then ON (ECU resets, glow)**
- Starter — **ENGAGE if propeller stopped (below 15,000 ft)**

▶ Engine restarted?

Yes:

- Power — **INCREASE slowly, monitor temperatures**
- ALTERNATOR — **ON**
- Deferred: Engine restarted in flight – land as soon as possible, maintenance inspection. — **NOTED**

No:

Secure the engine as below.

No – secure engine:

Securing the failed engine:

- Power lever (failed engine) — **IDLE**
- ENGINE MASTER (failed engine) — **OFF – propeller feathers automatically**
- Feathering — **CONFIRM (propeller stopped / low RPM)**
- ALTERNATOR (failed engine) — **OFF**
- Fuel selector (failed engine) — **OFF (or X-FEED to operating engine for balance)**
- Electrical load — **REDUCE (single alternator)**
- Bank / rudder — **5° into operating engine, ball half out, rudder trim**
- ATC — **DECLARE, request nearest suitable airport**

Deferred: One engine inoperative – single-engine approach, Vyse 85 KIAS, gear and full flap only when landing assured, go-around not possible below approx 4

One-Engine-Inoperative Approach & Go-Around

One engine secured and feathered

- Fuel selector (operating engine) — **NORMAL or X-FEED as required**
- Airspeed — **85 KIAS (Vyse) minimum until landing assured**
- Flaps — **APP when established on final (below 137 KIAS)**
- Landing gear — **DOWN – three green when landing assured (approx 500 ft AGL)**
- Flaps — **LDG only when landing assured (below 113 KIAS)**
- Airspeed — **FINAL 86 KIAS, reduce power lever gradually**
- Rudder trim — **NEUTRAL before touchdown**

▶ Go-around required?

Yes – above approx 400 ft AGL:

- Power lever (operating engine) — **MAX**
- Flaps — **APP, then UP above 85 KIAS**
- Landing gear — **UP (positive rate)**
- Airspeed — **85 KIAS (Vyse), bank 5° into operating engine**

Single-engine go-around below approx 400 ft AGL or with gear down and full flaps may be impossible – commit to land.

No – continue:

- Landing roll — **EXPECT asymmetric braking / yaw, keep straight with rudder and brake**

Engine Fire in Flight

LAND IMMEDIATELY

- Power lever (affected engine) (memory) — **IDLE**
- ENGINE MASTER (affected engine) (memory) — **OFF – propeller feathers**
- Fuel selector (affected engine) (memory) — **OFF**
- ALTERNATOR (affected engine) (memory) — **OFF**
- Cabin heat / defrost (memory) — **OFF**
- Airspeed — **INCREASE if fire persists (sideslip away from fire)**
- Landing — **ONE-ENGINE-INOPERATIVE APPROACH, nearest suitable field (do not restart)**

Engine Fire During Start

- ENGINE MASTER (affected engine) (memory) ..... **OFF**
- Fuel selector (affected engine) (memory) ..... **OFF**

ENGINE MASTER (other engine) ..... **OFF**

Electric master ..... **OFF**

Fire extinguisher ..... **OBTAIN**

Aircraft ..... **EVACUATE**

Cabin / Electrical Fire

LAND AS SOON AS POSSIBLE

- Electric master (memory) ..... **OFF**
- Avionic master (memory) ..... **OFF**

Cabin heat / fresh air ..... **CLOSE**

Fire extinguisher ..... **ACTIVATE**

Engines keep running: each ECU has its own backup battery when the electric master is off. Gear and flaps are electric/hydraulic – plan an emergency (free-fall) gear extension.

▶ Fire out and electrical power required?

Yes:

Circuit breakers ..... **CHECK – do not reset tripped breakers**

Electric master ..... **ON**

Avionics / loads ..... **ON one at a time, check for smoke**

Deferred: Electrical fire – suspect circuits isolated, affected systems unavailable. ..... **NOTED**

No:

Continue with electrics off. Standby instruments only, no flaps, gear by emergency extension.

Deferred: All electrics off – emergency gear extension, flaps-up landing (approx 95 KIAS), no radios. ..... **NOTED**

Emergency Descent

- Power levers (memory) ..... **IDLE**
- Flaps (memory) ..... **UP**
- Landing gear (memory) ..... **DOWN (below 194 KIAS) to steepen descent**
- Airspeed (memory) ..... **194 KIAS smooth air / 155 KIAS turbulence**

Emergency Gear Extension (free-fall) / Gear-Up Landing

Gear does not extend normally or hydraulic / electrical failure

Airspeed ..... **REDUCE below 156 KIAS**

GEAR (hydraulic pump) circuit breaker ..... **PULL**

Gear selector ..... **DOWN**

Emergency gear extension handle ..... **PULL fully and HOLD until three green (gear free-falls, approx 30 s)**

Landing gear ..... **CHECK three green (test light bulbs if not)**

▶ Three green?

Yes:

Yaw / sideslip ..... **ASSIST lock if needed**

Deferred: Gear extended by emergency system – do not retract, hydraulic pump CB left out, normal landing, maintenance inspection. ..... **NOTED**

No – gear-up / unsafe landing:

Airport ..... **SELECT longest paved runway, emergency services alerted**

Fuel ..... **BURN DOWN to minimum practical, fuel transfer pumps OFF**

Seats / belts / loose objects ..... **SECURE**

Flaps ..... **LDG**

Approach ..... **NORMAL 86 KIAS, minimum rate of descent at touchdown**

Front canopy ..... **UNLOCK to cooling gap position before touchdown**

ENGINE MASTERS ..... **OFF just before touchdown (propellers feather)**

Fuel selectors ..... **OFF**

Electric master ..... **OFF after touchdown**

Aircraft ..... **EVACUATE when stopped**

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